

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2087

Comanche Peak 138 kV
April 23, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2087. The Interconnection Customer proposes a 120.3 MW solar photovoltaic facility interconnecting at the Comanche Peak 138 kV substation in the Virginia area. The request was received on May 19, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 725 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,136,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Cowlitz Falls-Saranac Facility 138 KV line is overloaded to 139.7% of its 221 MVA emergency rating for the loss of the Cia-New Sterling 138 KV. The project contributes 7.8% DFAX (9.4 MW) to this overload.

For the loss of S Houston-Dessau 138 KV, the Clark-Comanche Peak 138 KV L loads to 135.5% of its 280 MVA rating. Project DFAX: 37.2%.

For the loss of Cowlitz Falls-Cia 138 KV, the Cia-New Sterling 138 KV L loads to 133.7% of its 408 MVA rating. Project DFAX: 12.8%.

Mottman-Bloomington 138 KV Line 2: 132.7% loading (476 MVA) following the outage of the Mottman-Bloomington 138 KV Cir. 1; DFAX = 23.3%.

Cowlitz Falls-Cia 138 KV L: 129.9% loading (308 MVA) following the outage of the New Sterling-Cia Substation 138 KV; DFAX = 7.8%.

The Clark-Booneville 138 KV L is overloaded to 129.5% of its 262 MVA emergency rating for the loss of the South Houston-Dessau 138 KV L. The project contributes 37.2% DFAX (44.7 MW) to this overload.

For the loss of Dessau-South Houston 138 KV, the Bloomington-Booneville Substation 138 KV loads to 127.9% of its 233 MVA rating. Project DFAX: 37.2%.

The Bloomington-Mottman 138 KV #1 is overloaded to 125.6% of its 503 MVA emergency rating for the loss of the Mottman-Bloomington 138 KV Ckt 2. The project contributes 23.3% DFAX (28.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cowlitz Falls-Saranac Facility 138 KV	Rebuild line with higher-capacity conductor	\$693,000	37% (\$255,000)
Clark Substation-Comanche Peak 138 KV	Rebuild line with higher-capacity conductor	\$2,174,000	57% (\$1,250,000)
New Sterling-Cia 138 KV	Rebuild line with higher-capacity conductor	\$1,579,000	28% (\$444,000)
Bloomington-Mottman 138 KV L	Rebuild line with higher-capacity conductor	\$1,164,000	39% (\$457,000)
Cowlitz Falls-Cia 138 KV L	Rebuild line with higher-capacity conductor	\$2,588,000	37% (\$954,000)
Clark-Booneville 138 KV L	Rebuild line with higher-capacity conductor	\$7,498,000	57% (\$4,312,000)
Bloomington-Booneville Substation 138 KV	Rebuild line with higher-capacity conductor	\$3,257,000	57% (\$1,873,000)
Bloomington-Mottman 138 KV #1	Rebuild line with higher-capacity conductor	\$1,506,000	39% (\$591,000)
	Total allocated to project		\$10,136,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.